

PROJECT INITIATION DOCUMENT (PID)

Campus Complaint Management System (CCMS)

Version: 1.0

Duration: 3 Weeks

Technology Stack:

- ReactJS
 - Node.js
 - Express.js
 - MongoDB
 - Git & GitHub
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1. PROJECT OVERVIEW

Background

In colleges, students often face issues related to classrooms, laboratories, hostels, libraries, internet connectivity, electrical facilities, and other campus infrastructure. Complaints are usually communicated through informal channels, making it difficult to track their status and resolution.

The Campus Complaint Management System (CCMS) provides a centralized platform where students can raise complaints, track their status, and receive updates while administrators can manage and resolve complaints efficiently.

2. PROJECT OBJECTIVES

The objectives of the project are:

- Develop a web application using the MERN Stack.
 - Provide an online complaint registration system.
 - Enable students to track complaint status.
 - Allow administrators to manage complaints.
 - Practice frontend, backend, database, and integration concepts.
 - Simulate Agile-based project development.
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3. PROJECT SCOPE

In Scope

Student Features

- Register/Login
- Submit Complaint
- View Submitted Complaints
- Track Complaint Status

Administrator Features

- View Complaints
- Update Complaint Status
- Resolve Complaints
- View Complaint Statistics

Dashboard Features

- Total Complaints

- Pending Complaints
 - Resolved Complaints
 - Category-wise Complaints
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Out of Scope

The following features are not required:

- SMS Alerts
- Mobile Application
- Multi-level Approval Workflow
- File Uploads
- Chat Support

4. USER ROLES

Student

Responsibilities:

- Submit complaints
- View complaint history
- Track complaint progress

Administrator

Responsibilities:

- View complaints
- Update complaint status
- Mark complaints as resolved

5. FUNCTIONAL REQUIREMENTS

Module 1: Authentication

Features

- User Registration
- User Login
- Logout

Fields

Users:

- Name
- Email
- Password
- Role

Role Values:

- Student
 - Admin
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Module 2: Complaint Management

Features

- Add Complaint
- Edit Complaint (Before Review)
- Delete Complaint

- View Complaint List

Complaint Fields

Field	Description
Complaint ID	Auto Generated
Complaint Title	Short Title
Category	Complaint Type
Description	Detailed Complaint
Location	Area of Issue
Date Submitted	Auto Generated
Status	Pending/In Progress/Resolved

Complaint Categories

Students can select:

- Classroom
- Laboratory
- Hostel
- Library
- Internet/Wi-Fi
- Electrical
- Water Supply

- Cleanliness
 - Other
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Module 3: Complaint Tracking

Features

- View My Complaints
- Track Current Status
- View Resolution Details

Status Values

- Pending
 - In Progress
 - Resolved
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Module 4: Admin Complaint Handling

Features

- View All Complaints
- Filter Complaints
- Update Status
- Mark Resolved

Filters

- Category
- Status
- Date

Module 5: Dashboard

Student Dashboard

Display:

- Total Complaints Raised
- Pending Complaints
- Resolved Complaints

Admin Dashboard

Display:

- Total Complaints
 - Pending Complaints
 - In Progress Complaints
 - Resolved Complaints
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6. NON-FUNCTIONAL REQUIREMENTS

Performance

- Application should load within 3 seconds.

Security

- Password encryption using bcrypt.
- Protected routes.

Usability

- Responsive UI.
- Easy navigation.

Reliability

- Data should persist in MongoDB.
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7. DATABASE DESIGN

Collection: Users

Fields:

- _id
 - name
 - email
 - password
 - role
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Collection: Complaints

Fields:

- _id
- title
- category
- description
- location
- status
- createdBy

- createdDate
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8. TEAM STRUCTURE

Team Lead

Responsibilities:

- Sprint Planning
- Task Allocation
- GitHub Management
- Code Integration
- Deployment Support

Feature Ownership:

- Dashboard Module
 - Project Integration
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Member 1 – Authentication Module

Responsibilities:

Frontend:

- Registration Page
- Login Page

Backend:

- Register API

- Login API

Database:

- User Collection

Deliverable:

- Complete Authentication Module
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Member 2 – Complaint Submission Module

Responsibilities:

Frontend:

- Complaint Form
- Complaint Listing

Backend:

- Create Complaint API
- View Complaint API

Database:

- Complaint Collection

Deliverable:

- Complaint Submission Module
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Member 3 – Complaint Tracking Module

Responsibilities:

Frontend:

- My Complaints Page
- Complaint Details Page

Backend:

- Track Complaint API
- Complaint Status API

Deliverable:

- Complaint Tracking Module
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Member 4 – Admin Management Module

Responsibilities:

Frontend:

- Admin Complaint List
- Status Update Screen

Backend:

- Update Status API
- Filter API

Deliverable:

- Complaint Management Module
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Team Lead / Member 5 – Dashboard Module

Responsibilities:

Frontend:

- Dashboard Cards
- Statistics Page

Backend:

- Dashboard APIs

Deliverable:

- Dashboard Module
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9. SPRINT PLAN

Sprint 1 – Frontend Development (Week 1)

Deliverables:

- React Project Setup
- Routing
- UI Screens
- Forms
- Navigation

Expected Output:

Working frontend with dummy data.

Sprint 2 – Backend Development (Week 2)

Deliverables:

- MongoDB Setup
- Express APIs
- CRUD Operations
- Postman Testing

Expected Output:

Working backend APIs.

Sprint 3 – Integration & Testing (Week 3)

Deliverables:

- Axios Integration
- End-to-End Testing
- Bug Fixing
- Final Demonstration

Expected Output:

Fully Functional MERN Application.

10. GITHUB WORKFLOW

Branch Structure:

main

development

feature/authentication

feature/complaints

feature/tracking

feature/admin

feature/dashboard

11. ACCEPTANCE CRITERIA

The project will be accepted when:

- Authentication is functional.
 - Complaint submission works.
 - Complaint status updates work.
 - Dashboard displays correct statistics.
 - Frontend and backend are integrated.
 - MongoDB stores and retrieves data successfully.
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12. FINAL DELIVERABLES

Each Team Must Submit:

1. GitHub Repository
 2. Project Report
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SUCCESS CRITERIA

Students should demonstrate:

- React Components and Routing
- Express API Development
- MongoDB CRUD Operations
- Axios Integration
- Team Collaboration
- GitHub Usage
- Basic Agile Development Process